

Long Nguyen Khac

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Education

International School, Vietnam National University, Hanoi

Hanoi, Vietnam

B.Eng. in Automation and Informatics

Sep. 2021 – Dec. 2026

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore (05/2025 – 05/2026).
- **Research interest:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, Nanyang Technological University

Singapore

Research Fellow

Feb. 2026 – Mar. 2026

- Conducted research on relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configurations, power transfer efficiency, and system-level integration for docking-free underwater charging.
- Explored mechanical and communication requirements for integrating wireless charging into an AUV prototype.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Taiwan / Remote

Research Intern

May 2025 – Present

- Applying machine learning models, including Random Forest, XGBoost, and Support Vector Regression, to predict cutting forces in CNC milling from Taguchi machining datasets.
- Developed data-driven modeling approaches for cutting force prediction in CNC milling using experimental datasets.
- Applied nonlinear system analysis, sparse identification, and machine learning methods to model machining dynamics.
- Built data processing and model evaluation pipelines for machining force prediction under different experimental conditions.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS

Hanoi, Vietnam

Student Researcher

Sep. 2022 – Present

- Conducting research on adaptive control for AUVs, focusing on station keeping for planar wireless charging.
- Designed the mechanical model and physical prototype of a 3-DOF delta parallel robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Built a dataset of 2,632 construction-site images and trained YOLOv5s-based models for personal protective equipment detection.

Publications

1. **Long Nguyen Khac** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot*. International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: [10.1007/978-3-031-80943-9_57](https://doi.org/10.1007/978-3-031-80943-9_57).
 - **Main highlights:** Adaptive fuzzy backstepping control for robust trajectory tracking of a 3-DOF delta parallel robot.
 - **Presentation:** Mechatronics and Automation section, The 3rd International Conference ICTA 2024.
2. **Long Nguyen Khac** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites*. Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
 - **Main highlights:** YOLOv5s-based PPE detection using six activation functions and a dataset of 2,632 construction-site images.
 - **Presentation:** National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

Selected Honors & Awards

Most Popular Poster Award , NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Mar. 2026
NUS ASEAN SPARK Fellow , NUS Enterprise Summer Programme in Entrepreneurship	Jul. 2026
Outstanding Young Face Award , Vietnam National University, International School, awarded in 2024 and 2025	May 2025
TOSHIBA Scholarship , TOSHIBA International Foundation	Sep. 2024
Fighting Spirit Prize , The IEEE Southeast Asia Circuits and Systems Society Hackathon	Dec. 2022
Merit-based University Academic Scholarship , Six-semester recipient for excellent academic performance	2021–2025

Selected International Experience

Nanyang Technological University

Singapore

Global Connect Fellow

Feb. 2026 – Mar. 2026

- Selected as 1 of 39 young scholars worldwide and 1 of 5 outstanding Vietnamese young scholars for a fully funded in the Global Connect Fellowship program at NTU, Singapore (~2% acceptance rate from ~2,700 applicants).
- Joined an leading research environment with exposure to interdisciplinary work across energy systems, and robotics.

National University of Singapore

Singapore

Exchange Student

Aug. 2025 – Dec. 2025

- Selected as one of 8 outstanding VNU students receiving a fully funded scholarship DiscoverNUS & the NUS ASEAN Master's Scholarship Programme for one semester exchange in NUS.
- Completed a one-semester academic exchange, gaining broader exposure to engineering education, international research culture, and interdisciplinary learning.

- Selected as one of 28 global students to intern at National Chung Cheng University funded by Taiwan Experience Education Program (TEEP), Ministry of Education, Taiwan
- Worked at Advanced Institute of Manufacturing with High-tech Innovations (AIM-HI) – one of leading tech institutions in Taiwan, participating in smart sensing manufacturing research project.

ASEAN in Today's World, Kyushu University

Indonesia

VNU representative

Feb. 2025

- Selected as one of four excellent VNU representatives to participate the program funded by Kyushu University, Japan.
- Participated in an ASEAN-focused academic program, strengthening regional understanding, cross-cultural communication, and awareness of science, technology, and society in Southeast Asia.

Temasek Foundation Specialist's Community Action & Leadership Exchange

Singapore

VNU Representative

Feb. 2024

- Selected as one of 29 VNU students to receive a scholarship for the program.
- Contributed to a group project on youth leadership and sustainability in urban development in Vietnam and Singapore.
- Engaged in a regional leadership and community action program, developing teamwork, communication, and social responsibility in an international environment.

Selected Teaching Experience**Faculty of Engineering and Technology, VNUIS**

Hanoi, Vietnam

Teaching Assistant

Sep. 2023 – Dec. 2024

- Modeling and Simulation of Control Systems**
 - Assisted 30 students across 7 laboratory sessions on MATLAB/Simulink for robotics, mechatronics, and process control applications.
 - Supported the preparation of laboratory manuals, tutorial exercises, and simulation-based learning materials.
- Microprocessors and Microcontrollers**
 - Mentored 30 students through 15 laboratory sessions on Embedded C, Arduino, STM32, and embedded system development.

Selected Working Experience**EOTEK Automation Solutions JSC**

Hanoi, Vietnam

Engineering Intern

Mar. 2025 – May 2025

- Project:** AGV Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
 - Sketched AGV/AMR traffic intersections using AutoCAD and simulated warehouse logistics operations using Visual Components Premium 4.6.
 - Monitored AGV/AMR behavior, identified simulation errors, and supported troubleshooting of automated logistics workflows.

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade

Hanoi, Vietnam

Collaborator

Jul. 2023 – Sep. 2023

- Project:** Consultancy Support and Sales Policy Development for E-Commerce Website.
 - Coordinated with SMEs across Dak Nong and An Giang provinces to support e-commerce platform development.
 - Assisted in website design, content management, and preparation of technical manual documents.

Skills

Research	Mathematical modeling, control systems design, autonomous systems (AUV, mobile robots), wireless power transfer, data-driven modeling
Programming	MATLAB, Python, Embedded C/C++, PLC programming
Software	MATLAB/Simulink, SolidWorks, Altium, Proteus, Visual Components, TIA Portal
AI	Data-driven modeling (SINDy, SVD), Machine learning (Random Forest, XGBoost), Computer vision (YOLO)
Languages	Vietnamese (Native), English (Fluent), French (Beginner), Chinese (Beginner)

References**Asst. Prof. Yun Yang**

Assistant Professor, School of Electrical & Electronic Engineering, NTU

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Prof. Her-Terng Yau

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Assoc. Prof. Nguyen Nhu Tung

Senior Lecturer, Faculty of Engineering and Technology, VNUIS

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Dr. Le Xuan Hai

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